

Solving *La Ghigliottina*

with a Typed PPMI Word Graph

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Introduction and Motivations

La Ghigliottina (final round of *L'Eredità*): five clue words, find the one word that connects to all five.

The link is **not** semantic similarity, it is **collocational and cultural**: the answer forms a fixed multi-word expression, compound, or idiom with each clue.

legno · forte · continua · malattia · veloce $\Rightarrow$ cassa

cassa di legno · cassaforte · cassa continua · cassa malattia · cassa

Distributional models (word2vec/fastText) are a poor structural fit – the signal is syntagmatic, not paradigmatic.

Related Work (1)

The EVALITA NLP4FUN shared task framed *La Ghigliottina* as a computational language game. The seminal player *OTTHO* (Basile et al. 2009) solved it by *knowledge infusion*, retrieval / co-occurrence over a dictionary, proverb lists, and Wikipedia, and **beat learned-embedding methods**, the outcome we reproduce.

The best system, *Il Mago della Ghigliottina* / *UNIOR4NLP* (Sangati et al. 2018, 2020), same MWE hypothesis, same resources (PAISÀ, De Mauro *polirematiche*, proverbs, compounds), reaches **61.0%** (2018) / **68.6%** (2020). This project set out to replicate and extend it, but fell short: our reproduction hits only **0.31**, and no extension moved acc@1.

Related Work (2)

Our contribution is not a new model but a controlled ablation showing why the shortfall is a coverage gap, not a scoring one:

- Their edge is broader curated resources (itWaC, Wikipedia titles, larger MWE/proverb lists).
- On 30k entities, knowledge-graph embeddings are dominated by **hubness**.
- Every recall-broadening extension trades top-1 accuracy for rank-tail recall.
- The signal is **syntagmatic**, so word2vec / fastText are a poor structural fit.

Proposed Approach

1. **Typed, weighted word graph** from PAISÀ. Edge *types* are MWE-bearing patterns – MOD (noun–adj), BINOMIAL (X e Y), DI (X di Y), DA (X da Y). Edge *weights* are **pattern-conditioned PPMI** (PMI within each relation’s own pairs).
2. **Curated IDIOM edges** from Italian *polirematiche* + proverbs ($\approx 18.5\text{k}$ pairs). They supply the clue→answer links running text can’t – added at a constant weight.
3. **Flat PPMI-sum solver**. Per clue take the strongest edge to a candidate (max over relations/directions), then sum across the five clues. Highest sum wins.
4. *Optional* **description pass**. A local Qwen3-1.7B explains the answer, one sentence per clue, post-hoc, plays no part in solving.

Main result

Full PAISÀ graph (600k triples / 32k entities), EVALITA 2018 test (100 games).

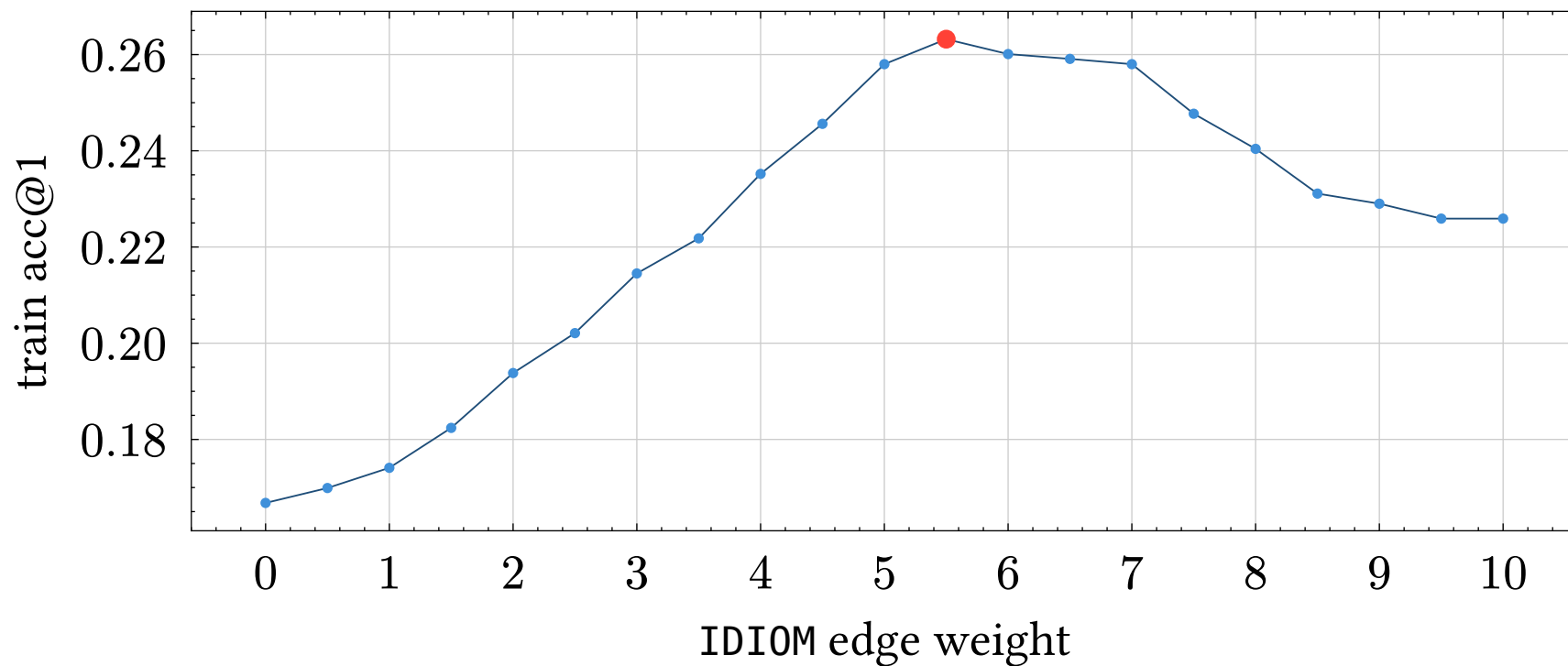
CONFIGURATION	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
flat-PPMI(sum), PAISÀ only	60	0.288	0.240	0.330	0.380	0.600
+ IDIOM (w=5.5)	67	0.384	0.310	0.480	0.520	0.670
IDIOM alone (no corpus)	43	0.206	0.160	0.240	0.270	0.430

Curated idiom edges are the **single largest lever** in the project: acc@1 0.24 → **0.31** (+29% rel), MRR 0.29 → 0.38.

Tuning the idiom weight (1)

The idiom edge weight is the one free scalar tuned on the **train** split (965 games, `tools/idiom_tune.py`), then held fixed on test. By plotting the idiom edge weight against `acc@1`, we get an inverted-U, with a peak **`acc@1 = 0.263` at $w = 5.5$** .

Tuning the idiom weight (2)



Description quality

The optional --describe explanations (Qwen3-1.7B) scored vs. the gold desc field (100 test games), generated for the solver's own top-1 prediction:

BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	BERTScore
1.70	0.248	0.035	0.177	0.170	0.658

BLEU on the sacreBLEU 0–100 scale; the rest 0–1.

The BLEU, ROUGE-1, ROUGE-2, and ROUGE-L scores are low, showing a low n -gram overlap. This is to be expected, as the generated text is free-form Italian, but the BERTScore shows the explanations are semantically on-topic.

Explored Extensions and Negative Results

Every extension improves recall / MRR / $\text{acc}@_{10,100}$, but **none improves $\text{acc}@1$** .

Models / scorers

- KGE (ComplEx, RotatE) – $\text{acc}@1$ 0.04 (hubness)
- Personalised PageRank – $\text{acc}@1$ 0.05
- Coverage / logsum scorers
- Learned re-ranker (10 features)

More data / more edges

- Wikipedia, itWaC joins (register miss)
- OpenSubtitles (helps recall, not top-1)
- COMPOUND, SIM (fastText) edges
- Wiktionary MWEs, ConceptNet

8 independent experiments · all confirm the same ceiling

Knowledge-graph embeddings – decisive loss

The project's original thesis: learned embeddings could score *unobserved* clue→answer links a graph baseline can't.

METHOD	FOUND	MRR	ACC@1
flat-PPMI(sum)	60	0.288	0.240
KGE ComplEx (300 ep)	21	0.023	0.010
KGE RotatE (150 ep)	37	0.068	0.040

Over 30k entities, link prediction is dominated by **hubness** – frequent adjectives / prepositions absorb probability mass, so the model smears its score instead of pinning the collocate (known partners rank #66–170).

Additional corpora – register beats volume

Extra corpora joined into the PPMI scan (PAISÀ-defined candidates):

ADDED CORPUS	+TOKENS	ACC@1	ACC@100	VERDICT
– (PAISÀ only)	–	0.240	0.600	baseline
Wikipedia	+650M	0.240	0.610	recall only
OpenSubtitles 12M	+85M	0.270	0.630	helps
OpenSubtitles full	+766M	0.250	0.660	recall ↑, top-1 ↓
itWaC (web)	+93M	0.220	0.600	hurts

Dialogue (85M tokens) beats Wikipedia (which doubled the graph). But nothing beats the curated idiom edges, the purest register match.

The recall-miss anatomy – why the ceiling exists

Gold-rank anatomy of the 100 test games:

RANK #1	RANK 2–10	RANK 11–100	MISSING (>100)
31	21	17	31

- The 31 misses **exist as graph nodes** but lack the clue→answer **edges** (98/100 gold already in vocab – coverage is not the problem).
- Training a re-ranker on the 965 train games does not help, it gave only 2 games difference on top-1.
- SIM / ConceptNet left train found essentially unchanged, it was the wrong kind of edge, not the missing syntagmatic collocations.

Conclusions and Limitations

A minimal system, typed pattern-PPMI over PAISÀ + curated idiom edges + flat sum, reaches **acc@1 0.31 / MRR 0.38** on EVALITA 2018 NLP4FUN, beating KGE, PageRank, and every corpus/edge extension.

The lesson: the task is bottlenecked by register-matched recall, not model sophistication. The one lever that ever moved top-1 was the idiom register.

Next step: a register-matched, associative phrase resource.

Thank you for your attention!